

Drone Noise Reduction through Audio Waveguiding

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ABSTRACT

Drones have become increasingly popular in today's world and extend to a wide variety of applications. However, one key limitation of drones even today is the audio noise emanating from their propellers. This noise is a characteristic of almost all drones available off-the-shelf and limits the capability of drones to record audio during flight. In this paper, we explore how simple techniques can help reduce and/or mitigate the drone propeller noise. Prior works have targeted addressing drone noise using sophisticated propeller designs and digital noise cancellation techniques. We aim at simple yet effective approaches that can be applied to any kind of drone without complex customization. One approach we broadly investigate in this paper is how to sound-proof the drone so as to absorb, cancel and/or redirect the audio noise away from the location of the microphone associated with the drone.

KEYWORDS

Drone; Noise; Movie Industry; Entertainment; Audio; Noise cancellation

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1 INTRODUCTION

The use of drones is moving at a rapid rate and we learn drone use cases that may not have been imagined before. Drones are increasingly being used in entertainment industry particularly for video footage for movie making. However, the setup when using drones for movie making, or aerial viewing in general, includes an embedded camera but not a microphone. This is due to the fact that an onboard microphone picks up the audio noise (high and low frequencies) at close range. It has infact been found through a NASA study [1] that most people find the noise from drones to be more annoying compared to that from heavy vehicular traffic on roads.

We encountered an experience of our own when we placed an Amazon Alexa voice assistant device in close proximity to a

Parrot minidrone. We observed that the smart voice device was not able to pick any sentences when the drone was switched ON and hovering close to the device. This motivated us to study the effect of noise on the device and cancel it in software using known signal processing techniques. However, as we continued our investigation we observed that the noise characteristics vary with each drone and type [2]. In this regard, we asked ourselves the question as to *if there is a simpler solution than characterizing the noise from different drones and digitally canceling the same?*

Motivated by the idea of sound-proofing in audio recording rooms, movie theaters and anechoic chamber constructions, in this paper we explore how sound proofing materials can be used effectively for *noise guiding* for drone applications. We refer to noise guiding instead of noise cancellation, as the key idea here due to the fact that the motive is to shield the recording device from the audio noise. This can be done by redirecting the noise through paths that are not in the reception range of the microphone.

In the rest of the paper we will discuss different preliminary design explorations for noise guiding on drones and our experiments to understand behavior of such designs. We will also present a brief overview of the next steps we aim to take in our goal towards noise guiding on drones.

2 DESIGN

The audio noise generated from drones is primarily due to its motors and propellers. Compared to the motors, propellers generate significantly higher amount of noise due to their high speed motion and thus drifting air. Drones work using the principle of aerodynamics. To lift-off from the propeller blades must spin at very high speeds. As these blades spin, it pushes the air close to the propellers downwards. This rapid contact of blades and air result in a chopping or buzzing sound, referred to as noise in human audible range. The motor contributes to the noise in terms of a whining tone but its intensity is significantly less compared to the propeller noise. Therefore, addressing drone noise primarily requires addressing its propeller noise.

Addressing propeller noise means the obvious solution is to design 'noiseless' propellers. While such designs are already available in the military and some custom use cases [3] the technology is very expensive and seems far from commercialization. We propose to design noise reduction solution where the noise can be guided or absorbed completely by a material add-on to the drone. In essence, we propose to fit an acoustic shroud that will shield the noise from drones through absorption and/or waveguiding in air.

Any add-on method to a drone must ensure that that it does not impede the flight performance of the drone. Clearly, it is possible to add heavyweight acoustic dampening materials on the drone, however, that can completely impede it from lift-off. Recently, a startup named Dotterel Technologies [4] have proposed to design

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an acoustic dampening shroud using nano-fiber materials. While this idea is promising, the cost of this material is very high and it also adds significant weight to the drone, thus limiting its use-case to larger drones. However, the fact that industry is also aiming at shroud based solution adds another motivating factor for our design.

Our approach aims to use inexpensive and lightweight materials as acoustic shrouds on drones for noise guiding. Our first choice for test materials is an acoustic dampening foam – typically used in anechoic sound proof chambers. This foam is extremely lightweight; a Parrot mini-drone can carry about 1ft x 1ft x 2in sheet of this foam. The next choice is aluminum foil (used in homes). Considering the wavelength of sound and the metallic nature of aluminum foil, the foil can act as a good sound reflector and redirecting agent. Our third choice, rubber (natural or synthetic polymer) is a good noise absorbing material. However, the key challenge with rubber is that its density is high and will add significant overhead weight to the drone. We believe that rubber as material is useful for noise guiding depending on with what material it is combined with – in which case a small amount of rubber is sufficient.

3 EXPERIMENTS AND RESULTS

We conducted preliminary experiments to understand the behavior of noise when using different shroud materials for noise waveguiding. We conducted experiments using a Parrot Mambo mini-drone. This drone is equipped with small but powerful motors, four 2" propeller blades and propeller guards. This drone is very noisy as its primarily designed as a racing drone. We placed the material shrouds over the propeller guards. Most drones have propeller guards and hence we felt this is an appropriate location on the drone for our preliminary tests. Figure 1, 2 3 shows the drone fit with foam, aluminum sheet and a combination of both, respectively.

The experiment setups in general are as shown in Figure 4. We conducted experiments by flying the drone and hovering it at a specific position relative to a microphone. The microphone is connected to a smartphone that records the audio as a .wav file. The audio intensity level (averaged over a duration of 30 sec) was also recorded using a audio decibel reading app on the smartphone.

In our initial experiment we also included the scenario where the microphone was placed on the drone (top and bottom). However, just by measuring the audio decibel level we observed that the noise level was very high and that the microphone was also picking up vibration noise; we confirmed this from our accelerometer readings by placing a phone on the drone resting on the floor with only the propellers ON. The experiment results are as shown in Figure 5.

In the next experiment, we set the distance between drone and microphone to 1ft and 3ft, and recorded the noise at various angles over a hemispherical region over the drone. We collected readings at 90, 45, 30, 0, -30, -45, and -90 degrees, where 0 refers to the microphone to the side of the drone at a specific distance (not on the drone). This experiment will help determine the localize the noise and how much of it is successfully blocked by the shrouds. Figures 6 and 7 show the noise level from the drone at these mic positions and for different shroud materials.

We make the following observations based on our preliminary experiments:



Figure 1: A Parrot mini-drone fit with the foam pads.



Figure 2: A Parrot mini-drone fit with the aluminum foil.



Figure 3: Drone fit with the foam and aluminum combo shroud.

- The best orientation, among our test locations of the mic, for least noise observance is at 0 degrees (to the side of the drone) and at 1ft distance. While larger distance helps reduce noise, a 1ft distance is realistic for mic placement on a drone as the add-on unit of 1ft length is practical as opposed to 3ft.
- Foam is the best material (compared to aluminum foil and rubber) for absorbing as it is light weight and easy to place on the drone. The foam was effective in absorbing the noise



Figure 4: Experiment setup

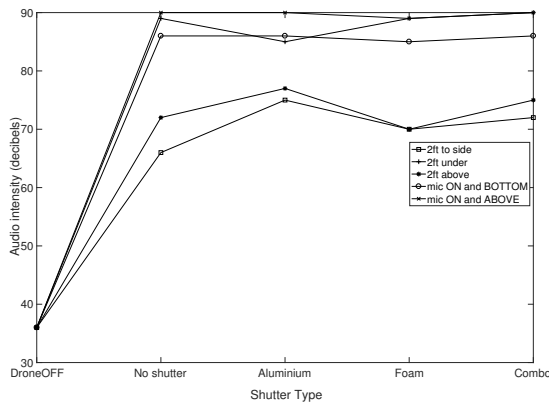


Figure 5: Preliminary investigations of audio intensity

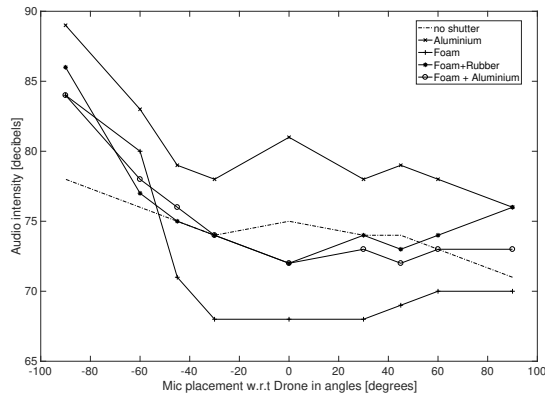


Figure 6: Audio intensity at 1ft

especially at 0 degrees. This reduced the noise by 4–6 decibels which is equivalent to going from heavy traffic noise to normal conversation from 3 feet distance.

- The aluminum shroud is ineffective in absorbing but reflects sound waves strongly. It is important to find an effective

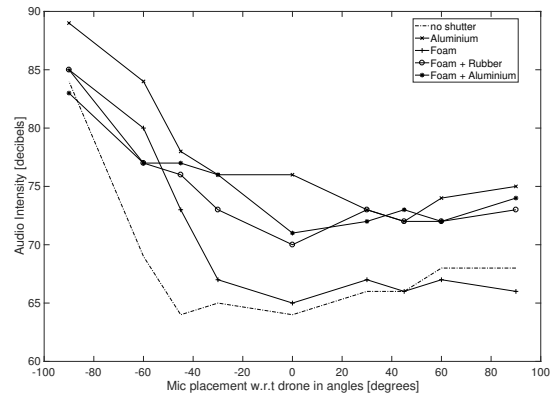


Figure 7: Audio intensity at 3ft

shape and placement for the shroud to create a strong wave-guiding effect.

- The rubber material is ineffective in the form of small flakes. However, rubber-like materials have good sound absorbing properties and thus we will continue to investigate compound rubber designs with foams to enhance the noise reduction.

4 CONCLUSION AND FUTURE WORK

This paper explored the idea of using sound absorbing and sound redirecting shrouds for drones to reduce audio noise pickup on embedded microphones. The design explored the use of foam, aluminum foil and rubber scrapes and a combination of those as potential shrouds. Preliminary experimental investigations reveal that a combination of foam and aluminum can help guide noise from drones. It was also observed that placing the microphone on the drone increases noise reception, however moving it away by 1ft along with use of shrouds can help reduce drone noise.

We aim to continue our investigations of noise reduction on drones. Motivated by the recent advancements in metamaterial structures to absorb/guide acoustic waves [5] we aim to 3D print simple lightweight structures that can fit onto drones and waveguide noise. In addition, we will also explore the use of polymer coatings and sheets (e.g. TPU films) as potential acoustic dampeners in our future designs.

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